



DATA SCIENCE & ANALYTICS PORTFOLIO

Three Projects in Predictive Modeling, Customer Analytics & Database Intelligence

Regression Modeling • Clustering & BI Dashboards • SQL Data Management



Profit Prediction



Customer Segmentation



Patient Records

What I'll Walk Through

Portfolio Overview



1. Profit Prediction

Multiple Linear Regression

Forecasts a startup's profit from R&D, admin & marketing spend.



2. Customer Segmentation

SQL + K-Means Clustering

Groups customers by behaviour using SQL, K-Means and Power BI.



3. Patient Records Analysis

SQL Database & Aggregation

Manages hospital data and surfaces insights via SQL queries.



PROJECT 1

Profit Prediction using Multiple Linear Regression



Problem solved: Predicts a startup's profit based on how it allocates spending across R&D, Administration and Marketing.

ALGORITHM

Multiple Linear Regression

A supervised machine learning algorithm that models the relationship between several input variables and one continuous output.

$$y = m_1x_1 + m_2x_2 + m_3x_3 + c$$

INPUTS

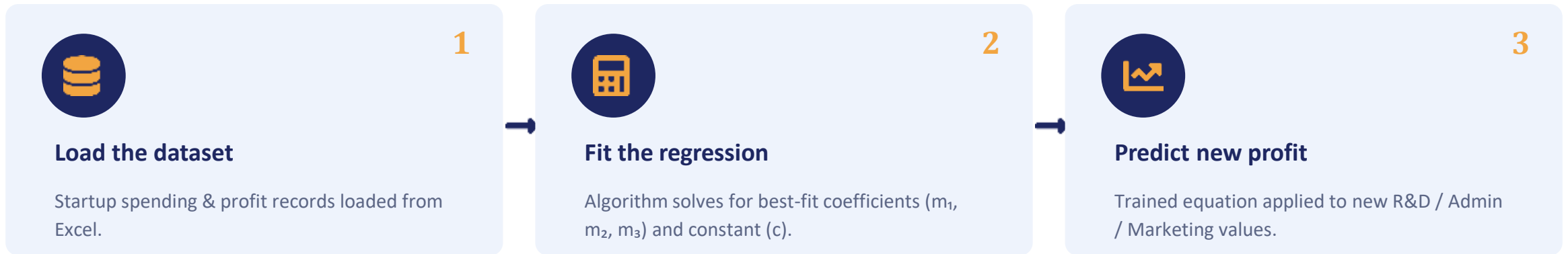
- R&D Spending
- Administration Spending
- Marketing Spending
- State



OUTPUT

Predicted Profit (₹)

From Raw Spending Data to a Profit Forecast



WORKED EXAMPLE

Predicted Profit for sample R&D, Admin & Marketing inputs:

≈ ₹79,037



VALUE DELIVERED

Helps businesses forecast profit ahead of time and make smarter, data-backed investment decisions across spending categories.

E74     f_x

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S				
1	RD_Spend	Administration	Marketing_Spend	State	Profit																		
2	165349.2	136897.8	471784.1	New York	192261.83				SUMMARY OUTPUT														
3	162597.7	151377.59	443898.53	California	191792.06																		
4	153441.51	101145.55	407934.54	Florida	191050.39				Regression Statistics														
5	144372.41	118671.85	383199.62	New York	182901.99				Multiple R	0.975062046													
6	142107.34	91391.77	366168.42	Florida	166187.94				R Square	0.950745994													
7	131876.9	99814.71	362861.36	New York	156991.12				Adjusted R Square	0.947533776													
8	134615.46	147198.87	127716.82	California	156122.51				Standard Error	9232.334837													
9	130298.13	145530.06	323876.68	Florida	155752.6				Observations	50													
10	120542.52	148718.95	311613.29	New York	152211.77				ANOVA														
11	123334.88	108679.17	304981.62	California	149759.96																		
12	101913.08	110594.11	229160.95	Florida	146121.95					df	SS	MS	F	Significance F									
13	100671.96	91790.61	249744.55	California	144259.4				Regression	3	75683964196	25227988065	295.9780624	4.52851E-30									
14	93863.75	127320.38	249839.44	Florida	141585.52				Residual	46	3920856301	85236006.54											
15	91992.39	135495.07	252664.93	California	134307.35				Total	49	79604820497												
16	119943.24	156547.42	256512.92	Florida	132602.65																		
17	114523.61	122616.84	261776.23	New York	129917.04					Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%						
18	78013.11	121597.55	264346.06	California	126992.93				Intercept	50122.19299	6572.352622	7.626217867	1.05738E-09	36892.73332	63351.65266	36892.73332	63351.65266						
19	94657.16	145077.58	282574.31	New York	125370.37				RD_Spend	0.80571505	0.04514727	17.84637376	2.63497E-22	0.714838309	0.89659179	0.714838309	0.89659179						
20	91749.16	114175.79	294919.57	Florida	124266.9				Administration	-0.026815968	0.05102878	-0.525506752	0.601755108	-0.129531575	0.075899638	-0.129531575	0.075899638						
21	86419.7	153514.11	0	New York	122776.86				Marketing_Spend	0.027228065	0.016451235	1.6550773	0.104716819	-0.005886553	0.060342682	-0.005886553	0.060342682						
22	76253.86	113867.3	298664.47	California	118474.03																		
23	78389.47	153773.43	299737.29	New York	111313.02																		
24	73994.56	122782.75	303319.26	Florida	110352.25																		
25	67532.53	105751.03	304768.73	Florida	108733.99																		
26	77044.01	99281.34	140574.81	New York	108552.04																		
27	64664.71	139553.16	137962.62	California	107404.34																		
28	75328.87	144135.98	134050.07	Florida	105733.54																		
29	72107.6	127864.55	353183.81	New York	105008.31																		
30	66051.52	182645.56	118148.2	Florida	103282.38																		
31	65605.48	153032.06	107138.38	New York	101004.64																		
32	61994.48	115641.28	91131.24	Florida	99937.59																		
33	61136.38	152701.92	88218.23	New York	97483.56																		

File Home Insert Draw Page Layout Formulas Data Review View Help

Clipboard Font Alignment Number Styles Cells Editing Add-ins ChatGPT Numerous.ai GPT for Excel Word Data Analysis

57										
58										
59										
60	Prediction Section									
61										
62	Intercept	50122.19299								
63	RD Coefficient	0.80571505								
64	Admin Coefficient	-0.026815968								
65	Marketing Coefficient	0.027228065								
66										
67										
68	R&D	Administration	Marketing	Predicted Profit						
69	21892.92	81910.77	164270.7	70037.90477						
70	23940.93	96489.63	137001.1	70554.57256	Profit	50122.2				
71					(R&D)	0.8057				
72					Admin	- 0.0268				
73					Marketing	0.0272				
74										
75										

Paste

Cut Copy

Format painter

Clipboard

Get data

Excel workbook

OneLake catalog

SQL Server

Enter data

Data

Dataverse

Recent sources

Transform data

Refresh data

Queries

New visual

Text box

More visuals

Insert

New visual calculation

New measure

Quick measure

Calculations

Sensitivity

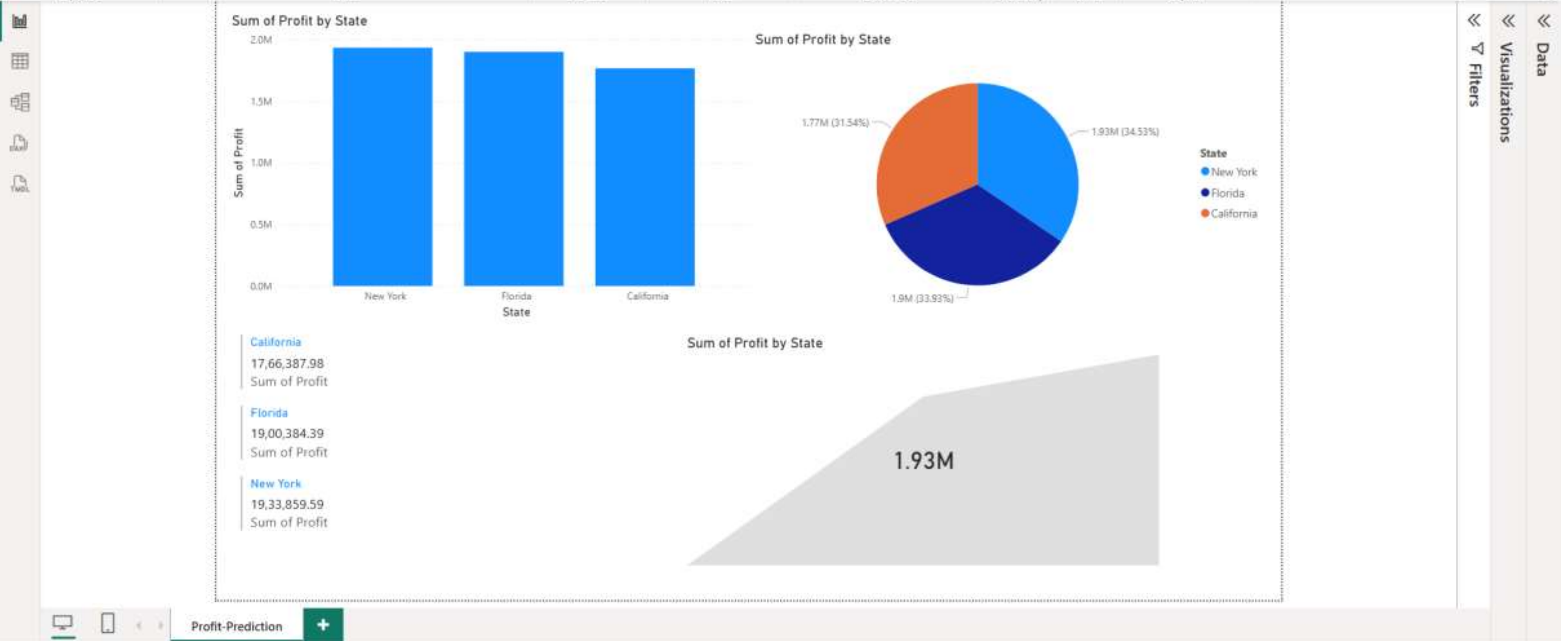
Sensitivity

Publish

Share

Prep data for Copilot AI

Copilot





PROJECT 2

Customer Segmentation using SQL + K-Means Clustering



Problem solved: Groups customers by shopping behaviour so marketing can be targeted to the right audience.

TOOLS USED



MySQL

Data cleaning & queries



Power BI

Segment dashboards



K-Means

Clustering ($k = 3$)

HOW IT WORKS

- 1 Cleaned & queried data in SQL — total sales, sales by category, by location, by payment method.
- 2 Applied K-Means clustering ($k = 3$) to group customers by age and spending.
- 3 Assigned each customer to the nearest cluster centroid, refining iteratively.

Three Distinct Customer Segments



Budget Customers

Low spending; buy frequently during discounts.



Premium Customers

High spending with frequent, high-value purchases.



Seasonal Customers

Spend spikes around festivals and sale periods.



OUTPUT

Interactive Power BI dashboards visualizing each customer segment.



VALUE DELIVERED

Helps businesses target the right customers, improve sales, and stabilize the customer base.



Navigator

SCHEMAS

Filter objects

- ▼ project_shopping_trends
 - ▼ Tables
 - shopping_table
 - shopping_table
 - std
 - ▼ Views
 - shopping
 - Stored Procedures
 - Functions

Administration Schemas

Information

No object selected

Object Info Session

SQL Editor Opened.

CustomerPurchase_SQL_Proc

Limit to 1000 rows

```
1  ## To Display all from table
2  • select * from shopping_table;
3
4  ## 1. This shows which payment methods customers use the most ##
5  • SELECT payment_method, COUNT(*)
6    FROM shopping_table
7    GROUP BY payment_method;
8
9  ## 2. See who spent the most moneyThis lists your customers ranked by their total spending
10 • SELECT customer_id, SUM(Purchase_Amount_USD)
11    FROM shopping_table
12    GROUP BY customer_id
13    ORDER BY SUM(Purchase_Amount_USD) DESC;
14
15  ## Find the average amount spent per payment typeThis helps you see if people spend more when using specific methods (like Credit Card v
16 • SELECT payment_method, AVG(Purchase_Amount_USD)
17    FROM shopping_table
18    GROUP BY payment_method;
19
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
---	------	--------	---------	------------------

SQLAdditions

Jump to

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help

Snippets



Filter...

Data

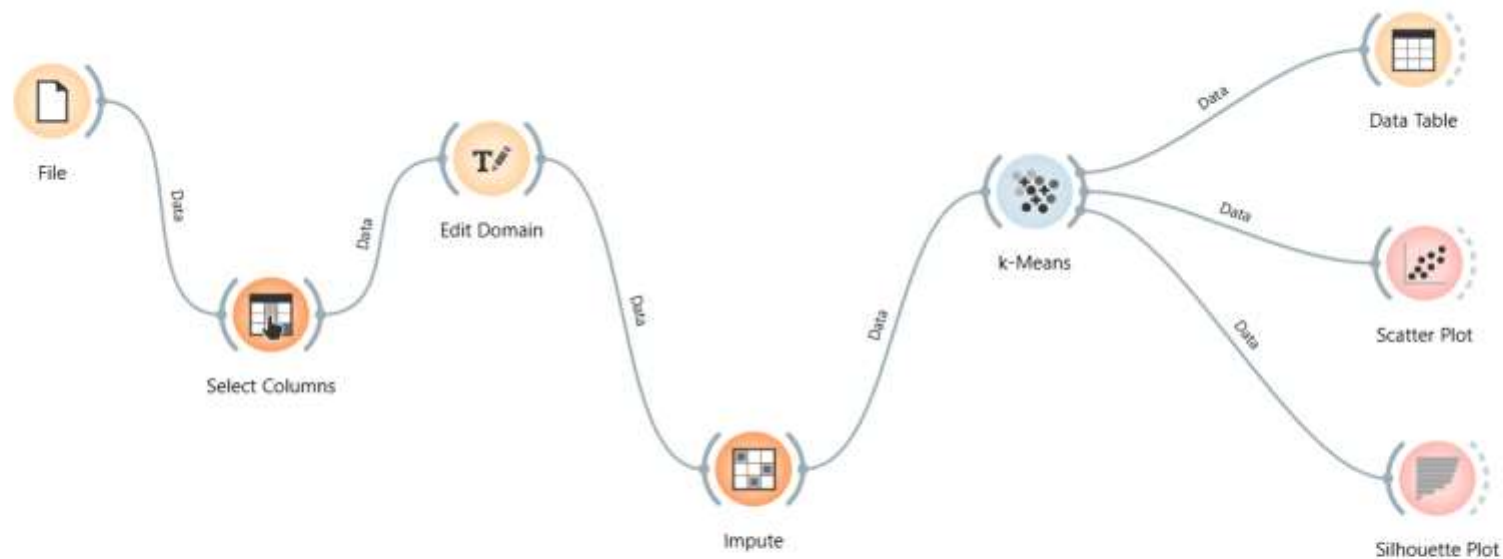
- File
- CSV File Import
- Datasets
- SQL Table
- Data Table
- Paint Data
- Data Info
- Rank
- Edit Domain
- Color
- Column Statistics
- Save Data

Transform

- Data Sampler
- Select Columns
- Select Rows
- Transpose
- Split
- Merge Data
- Concatenate
- Select by Data Index

Select a widget to show its description.

See [workflow examples](#), [YouTube tutorials](#), or open the [welcome screen](#).



Orange Data Mining is a **visual data analysis and machine learning software**.

It allows us to analyze data and create machine learning models by **connecting different tools (widgets)**, without writing much code.

Customer Segmentation in Orange

File → Upload the customer data.

Edit Domain → Change or correct column names/types if needed.

Select Columns → Select the columns we need for analysis.

Impute → Fill missing values in the data.

K-Means → Groups similar customers into different groups.

Data Table → Shows the customer data with their group/cluster.

Scatter Plot → Shows the customer groups visually.

Silhouette Plot → Checks how good the customer groups are.

Profit Analysis • Last saved: Today at 10:26 PM

Search

Sign in

Share

FileHomeInsertModelingViewOptimizeHelpFormatData / Drill

Clipboard

Get dataExcelOneLakeSQL ServerEnter dataDataverseRecent sources

Queries

Insert

Calculations

Sensitivity

Share

Copilot

Category	Sum of Purchase_Amount_USD
Accessories	18465
Clothing	26104
Footwear	9939
Outerwear	5145
Total	59653

Accessories

18465

Sum of Purchase_Amount_USD

Clothing

26104

Sum of Purchase_Amount_USD

Footwear

9939

Sum of Purchase_Amount_USD

Outerwear

Sum of Purchase_Amount_USD

Sum of Purchase_Amount_USD by Category

Category	Sum of Purchase_Amount_USD
Clothing	26104
Accessories	18465
Footwear	9939
Outerwear	5145

Sum of Review_Rating by Category

Category	Sum of Review_Rating
Clothing	1.61K (43.16%)
Accessories	1.19K (31.74%)
Footwear	0.33K (8.69%)
Outerwear	0.61K (16.41%)

Sum of Purchase_Amount_USD by Category

Category	Sum of Purchase_Amount_USD
Clothing	26K (43.76%)
Accessories	18K (30.95%)
Footwear	10K (16.66%)
Outerwear	3K (5.62%)

Sum of Purchase_Amount_USD by Payment_Method

Payment_Method	Sum of Purchase_Amount_USD
Credit Card	11K (43.21%)
Debit Card	10K (17.1%)
Venmo	10K (16.91%)
PayPal	10K (16.38%)
Cash	10K (16.2%)
Bank Transfer	9K (15.06%)

Filters

Visualizations

Data

Profit-Prediction

Page 1

Page 2 of 2

News for you
Sensex today | S...

Search

ENG IN

10:26 PM
22-09-2026



PROJECT 3

Medical Patient Records Analysis using SQL



Problem solved: Manages and analyzes hospital patient data efficiently, replacing manual record-keeping.



DATABASE

MedicalDB

A relational database created to store and manage structured hospital patient information in a single Patients table.

Table: Patients

Structured for fast filtering, sorting & aggregation

FIELDS STORED

Name

Age

Gender

Disease

Admission Date

Discharge Date

Treatment Cost

From Raw Records to Actionable Insight



Insert

Loaded patient records into the table.



Query

SELECT, WHERE, ORDER BY for targeted lookups.



Aggregate

COUNT, SUM, AVG, MAX, MIN across records.



Group

GROUP BY — e.g. avg. cost per disease.

EXAMPLE QUERIES

Highest / lowest treatment cost • Patients with a specific disease • Records sorted by age or admission date



VALUE DELIVERED

Helps hospitals quickly search, sort, and analyze patient data instead of relying on slow, error-prone manual record-keeping.

MySQL Workbench

project_medical_data_his

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

SCHEMAS

Filter objects

project_medical_data_history

- Tables
- Views
- Stored Procedures
- Functions

Administration

Schemas

Information

No object selected

Object InfoSession

Medical project

Limit to 1000 rows

1## Display all whats inside Table

2

3• SELECT * FROM patients;

4• SELECT * FROM admissions;

5• SELECT * FROM doctors;

6• SELECT * FROM province_names;

7

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	patient_id	first_name	last_name	gender	birth_date	city	province_id	allergies	height	weight
▶	1	Donald	Waterfield	M	1963-02-12	Barrie	ON	None	156	65
	2	Mickey	Baasha	M	1981-05-28	Dundas	ON	Sulfa	185	76
	3	Jiji	Sharma	M	1957-09-05	Hamilton	ON	Penicillin	194	106
	4	Blair	Diaz	M	1967-01-07	Hamilton	ON	None	191	104
	5	Charles	Wolfe	M	2017-11-19	Orillia	ON	Penicillin	47	10
	6	Sue	Falcon	F	2017-09-30	Ajax	ON	Penicillin	43	5
	7	Thomas	O'Neill	M	1993-01-31	Burlington	ON	None	180	117
	8	Sonny	Beckett	M	1952-12-11	Port Hawkesbury	NS	None	174	105
	9	Sister	Spitzer	F	1966-10-15	Toronto	ON	Penicillin	173	95
	10	Cedric	Coltrane	M	1961-11-10	Toronto	ON	None	157	61
	11	Hank	Spencer	M	1969-08-10	Peterborough	ON	None	158	74

patients 1

ApplyRevertContext HelpSnippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	21:32:51	SELECT * FROM patients LIMIT 0, 1000	1000 row(s) returned	0.109 sec / 0.047 sec

Query Completed

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

25°C Mostly cloudy

Search

ENG IN

09:36 PM 22-09-2026